Physical Chemistry CUME, Oct. 5th, 2018

Quantum Mechanics

Harmonic Oscillator Raising and Lowering operators

For a one-dimensional harmonic oscillator, it is possible to define the lowering and raising operators:

$\hat{a}_+ = \frac{1}{\sqrt{2\hbar m\omega}} \cdot (-i\hat{p} + m\omega\hat{x})$, and $\hat{a}_- = \frac{1}{\sqrt{2\hbar m\omega}} \cdot (i\hat{p} + m\omega\hat{x})$, where $\hat{a}_+$ is called the raising operator and $\hat{a}_-$ as the lowering operator.

If $|n>$ is an eigenstate of the Harmonic oscillator, $\hat{a}_+$ and $\hat{a}_-$ has the following properties,

$$\hat{a}_+ |n> = \sqrt{n+1}\,|n+1> ,$$

$$\hat{a}_- |n> = \sqrt{n}\,|n-1> ,$$

where the eigenstates with quantum numbers n+1 and n-1 are both properly normalized. (Recall for a normalized eigenstate $|n>$, we have $<n|n>=1$)

(a) Express the position operator $\hat{x}$ using the lowering and raising operators.

(b) Express the moment operator $\hat{p}$ using the lowering and raising operators.

(c) Calculate the expectation value for kinetic energy $\hat{K} = \frac{\hat{p}^2}{2m}$ for the ground state of the Harmonic oscillator $|0>$

(d) The energy of a true bond that is not completely harmonic can be estimated with the perturbation theory. Assume the real bond potential is $\frac{1}{2}m\omega^2 x^2 - cx^3$, where c is a small number. According to the perturbation theory, up to the 2nd order, the energy of the ground vibrational state of the bond is

$$\frac{1}{2}\hbar\omega - c^2 \sum_{n=1}^{\infty} \frac{<0|\hat{x}^3|n><n|\hat{x}^3|0>}{n \cdot \hbar\omega},$$

please evaluate the sum and write an simplified expression for the energy of the ground state.